

AgriVyaan

Smart Farming Assistant — Autonomous Drone Crop Inspection System

Smart India Hackathon 2026

Problem Statement ID: SIH26180

Organization: Qualcomm Inc.

Theme: Agriculture, FoodTech & Rural Development

Project Overview Report

Abstract

AgriVyaan is a low-cost, fully offline, drone-based crop inspection system built to bring precision agriculture within reach of India's smallholder and marginal farmers. Rather than requiring farmers to own or operate drones and sensors themselves, the system is delivered as a shared drone-as-a-service: a trained operator books, flies, and validates each scan, while the farmer simply receives a clear, actionable report on a mobile app — explained in their own language, even by voice.

The system performs a two-stage aerial inspection of potato fields — a fast, full-field survey followed by a targeted, close-up rescan of areas of concern — using an onboard Edge-AI pipeline that runs entirely on-device, with zero dependency on cloud connectivity or the internet. Findings such as disease, pest activity, water stress, and nutrient deficiency are verified by a human operator before ever reaching the farmer, ensuring every recommendation is transparent, evidence-based, and trustworthy.

AgriVyaan's core contribution is not simply a better crop-disease detector — it is a complete, trust-first adoption model designed around the real barriers that prevent Indian smallholder farmers from benefiting from precision agriculture today: low connectivity, low literacy, low trust in autonomous AI, and the inability to own expensive hardware.

Problem Statement

Smallholder and marginal farmers, who make up the vast majority of India's agricultural workforce, routinely lose crop yield to diseases, pests, water stress, and nutrient deficiencies that go undetected until visible symptoms have already spread across the field. By the time a problem is noticed with the naked eye, the opportunity for a low-cost, targeted intervention has often already passed.

Existing precision-agriculture and drone-based monitoring solutions largely assume conditions that do not hold for most Indian farmers:

- Reliable internet connectivity in the field, which is frequently unavailable in rural regions.
- Ownership of a drone, smartphone, or other precision-agriculture hardware, which is financially out of reach for small landholders.
- Comfort with reading and interpreting technical, text-heavy digital reports.
- Willingness to trust and act on fully autonomous AI decisions with no human verification.

A genuinely useful solution for this population has to be designed around these adoption barriers first — not just optimized for marginal gains in detection accuracy.

Proposed Solution

AgriVyaan addresses this as an access and trust problem as much as a detection problem, through a system built around three connected components:

1. Drone + Edge AI

A quadcopter drone autonomously surveys the field in two stages — a coarse full-field pass to flag areas of concern, followed by a close, high-resolution rescan of only those flagged zones. All image analysis, anomaly detection, and disease/pest recognition happens on-device in real time, with no cloud dependency. Flight itself remains fully autonomous throughout; no human intervenes mid-flight.

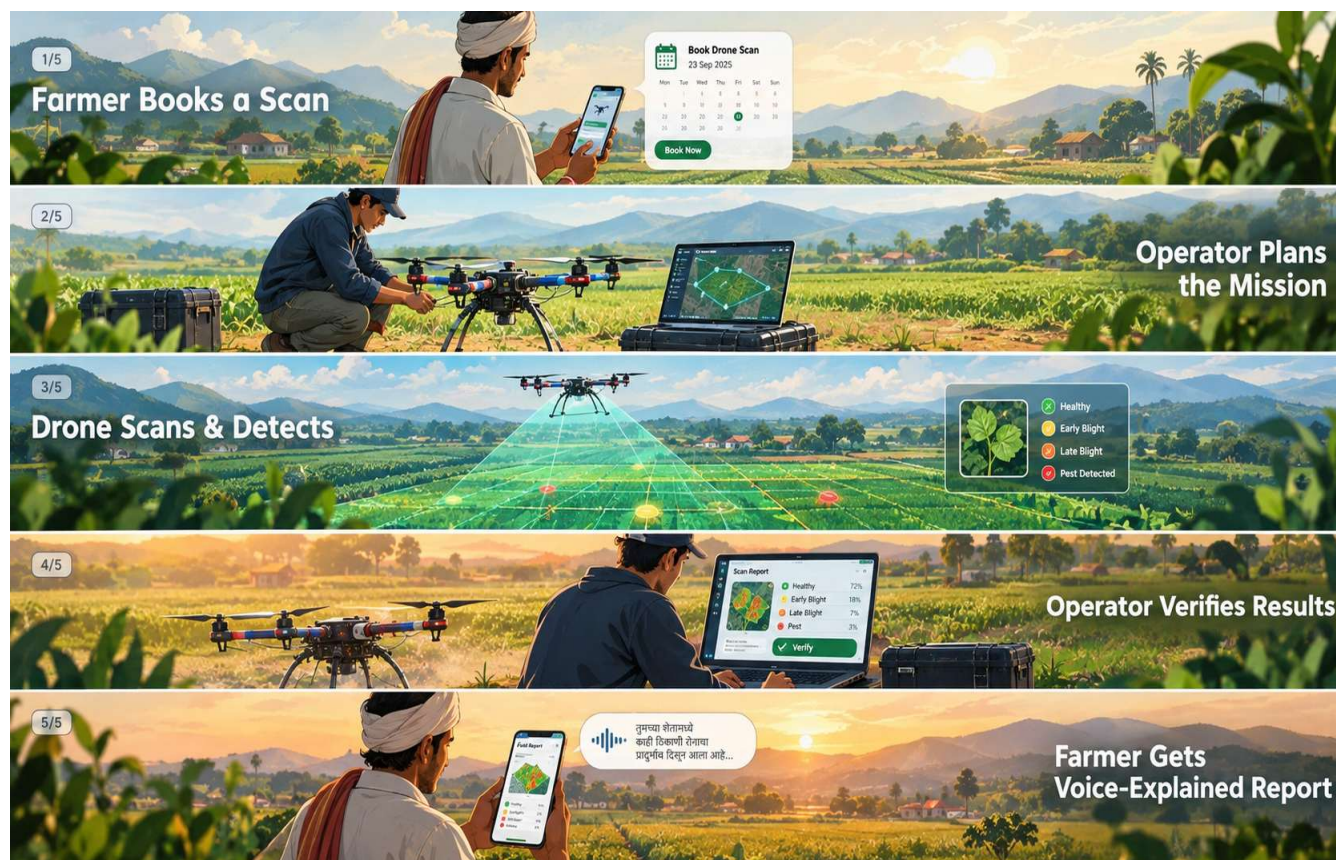
2. Operator Application

A field operator manages bookings, plans and supervises each mission, and reviews every AI-generated finding before it becomes part of a farmer's report. Ambiguous or uncertain findings are explicitly routed to the operator for a manual decision rather than being silently accepted or dismissed by the system, and any recommendation involving fertilizer or pesticide use requires operator sign-off before release.

3. Farmer Application (AgriVyaan)

An offline-first, multilingual mobile app that turns complex AI output into a simple, understandable report — colour-coded field zones, plain-language recommendations, and a voice assistant that can explain the report (and even guide first-time account setup) entirely by speech in the farmer's native language. This is designed specifically for farmers with limited digital literacy or limited comfort with text-heavy interfaces.

Together, these three layers form a complete loop: Detect → Verify → Explain → Recommend → Act → Track → Rescan, so that a finding doesn't just get reported — it gets acted on and its outcome gets followed up in a later scan.



System Overview

At a high level, the system architecture consists of three layers, each responsible for a distinct stage of the workflow:

Layer	Responsibility
Drone + Edge AI	Autonomous flight, on-device image capture and analysis, real-time anomaly and disease/pest detection, sensor fusion — fully offline.
Operator Application	Booking management, mission planning, flight supervision, human verification of AI findings, final report generation.
Farmer Application	Offline-first, multilingual report delivery, voice-guided explanation, recommended actions, product sourcing, and follow-up tracking.

The end-to-end journey moves from booking, to field survey, to targeted rescan of flagged areas, to human-verified reporting, to a farmer-facing action plan — and finally to a follow-up scan that confirms whether the recommended action actually improved the field.

Key Highlights

- Fully offline edge pipeline — the entire detect-to-report loop works with zero internet connectivity.
- Two-stage inspection — a fast full-field scan followed by a smart, targeted rescan for efficiency and accuracy.
- Human-verified findings — every recommendation is checked by a trained operator before it reaches the farmer.
- Voice-guided experience — onboarding and report explanation available in the farmer's native language.
- Autonomous ground-truth soil sampling — the drone can land safely to take a direct soil-moisture reading.
- Battery-aware, resumable missions — scans that exceed one flight's battery are queued and resumed, never dropped.
- Explainable by design — every finding is confidence-labeled and evidence-backed, never a black-box verdict.
- Closed-loop tracking — from problem detection through to recorded harvest outcome.

Impact & Benefits

Impact Area	Benefit
Reduced Crop Loss	Early, precise detection enables intervention before problems spread across the field.
Lower Input Costs	Zone-level findings support targeted treatment instead of blanket pesticide or fertilizer use.
Water Efficiency	Evidence-based irrigation guidance helps reduce both under- and over-watering.
Accessibility for Smallholders	A shared drone-as-a-service model removes the cost barrier of owning hardware.
Works Anywhere	Fully functional in low- or zero-connectivity rural regions.
Builds Trust	Confidence-labeled, human-verified findings support informed decisions rather than blind trust in AI.

Technology Stack

Drone & Edge AI

- Raspberry Pi (edge compute) · Pixhawk with ArduPilot (flight controller)
- Python, OpenCV, TensorFlow Lite / NCNN for on-device inference
- MAVLink protocol for Pi-flight-controller communication
- ESP32 + environmental sensor module for temperature and humidity

Operator Application

- React / Next.js front end
- Node.js backend services
- Local-first database with cloud sync (SQLite / PostgreSQL)

Farmer Application (AgriVyaan)

- React Native offline-first mobile app
- On-device OCR for Soil Health Card digitization
- Text-to-speech / speech-to-text for the multilingual voice assistant

Shared Infrastructure

- REST APIs connecting the operator and farmer applications
- Offline-first local storage with conflict-safe background synchronization

Conclusion

AgriVyaan demonstrates that precision agriculture does not need to wait for every farmer to own expensive hardware or have reliable internet. By combining a fully offline, autonomous drone pipeline with a human-verified reporting process and a voice-guided, multilingual farmer experience, the system is designed to fit the real conditions of Indian smallholder agriculture rather than assuming them away.

The result is a practical, trustworthy, and accessible path to earlier problem detection, lower input costs, and better-informed farming decisions — built around the people the system is meant to serve.